Problem Set 1: Solution

1. Impulse and step representations

The sequence x[n] is $x[-1] = 3$, $x[0] = -2$, $x[2] = 4$, and zero elsewhere.

a) Express $x[n]$ as a sum of scaled, shifted unit samples.

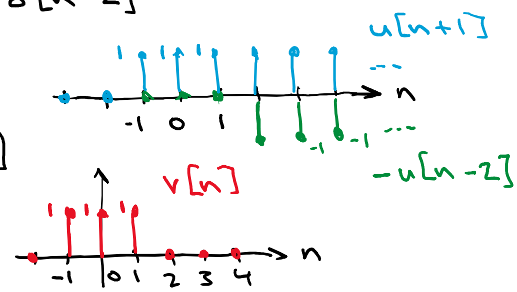
b) Let $v[n] = u[n+1] - u[n-2]$. List the non-zero samples of $v[n]$ and sketch it.

c) Express $\delta[n+2]$ using two shifted unit steps.

a) $x[n] = 3\,\delta[n+1] - 2\,\delta[n] + 4\,\delta[n-2]$

b) $v[n] = u[n+1] - u[n-2]$

$= \delta[n+1] + \delta[n] + \delta[n-1]$



c) $\delta[n+2] = u[n+2] - u[n+1]$

2. System-property classification

For each system below, determine whether it is linear, time-invariant, causal, and BIBO stable. Give a one- or two-sentence justification for every "no".

A. $y[n] = 2x[n-2]$

B. $y[n] = x^2[n]$

C. $y[n] = n\,x[n]$

D. $y[n] = x[n] + 0.5\,x[n-1]$

A. $y[n] = 2x[n-2]$ linear, time invariant, causal, BIBO stable
fixed gain and fixed delay

B. $y[n] = x^2[n]$ non-linear, time invariant, causal, BIBO stable
Squaring violates superposition
But $|x| \le B$ implies $|y| \le B^2$

C. $y[n] = n\,x[n]$ linear, time varying, causal
not BIBO stable

Bounded input $x[n] = 1$ produces $y[n] = n$
which is unbounded

D. $y[n] = x[n] + 0.5\,x[n-1]$ linear, time invariant, causal, BIBO stable
$h[n] = \delta[n] + 0.5\,\delta[n-1]$ and $\sum_n |h[n]| = 1.5 < \infty$

3. Finite convolution and support

Let $x[n] = \{1, -1, 2\}$ for $n = 0,1,2$ and $h[n] = \{2,1\}$ for $n = 0,1$.

a) Predict the index range over which $y[n] = x[n] * h[n]$ can be non-zero.

b) Compute the full convolution by hand.

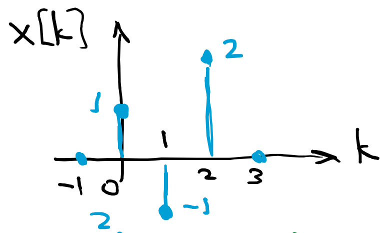
c) Verify the results with np.convolve.

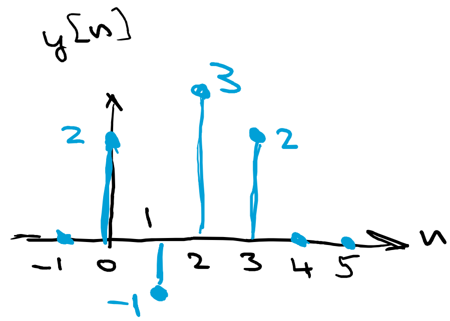
a) $x[n] = [1, -1, 2]$ $n=0$ $n=2$ $h[n] = [2, 1]$ $n=0$ $n=1$

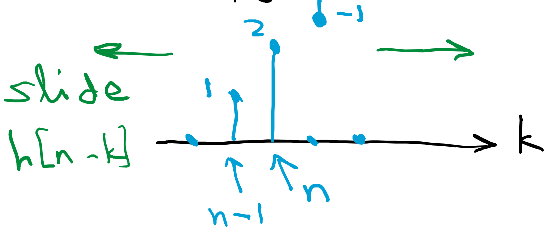
$$y[n] = \sum_k x[k] \cdot h[n-k]$$

$0 \le k \le 2$ $0 \le n-k \le 1$ or $k \le n \le k+1$

using bounds on k $0 \le n \le 3$ length 4








```
import numpy as np
```

← part c)

```
# Convolution using np.convolve
x = [1,-1,2]; h = [2,1]
y = np.convolve(x, h, mode='full')
print(f'x*h = {y}')
```

```
x*h = [ 2 -1  3  2]
```

```
import torch
import torch.nn.functional as F
```

```
# DSP convolution using torch conv1d
x2 = torch.tensor([[[0,1,-1,2,0]]])   # note the padding with zeros
w2 = torch.tensor([[[1,2]]])          # this is h[n] flipped
y2 = F.conv1d(x2, w2)
print(f'torch: x*w = {y2}')
```

```
torch: x*w = tensor([[[ 2, -1,  3,  2]]])
```

4. Impulse response, causality, stability, and step response

An LTI system has $h[n] = (0.75)^n\, u[n]$.

a) Is the system causal? Explain from $h[n]$.

b) Is the system BIBO stable? Show the relevant sum.

c) Find the response to $x[n] = u[n]$. Give a closed-form expression for $n >= 0$.

d) What is the limiting output value as $n \to \infty$?

a) $h[n] = (0.75)^n u[n] \Rightarrow h[n] = 0$ for $n < 0$

$\Rightarrow$ The system is causal

b) $\sum_n |h[n]| = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n = \dfrac{1}{1 - 3/4} = 4 < \infty$

$\Rightarrow$ BIBO stable

c) $y[n] = \sum_k h[k]\, x[n-k] = \sum_{k=0}^{n} \left(\frac{3}{4}\right)^n$

$= \dfrac{1 - \left(\frac{3}{4}\right)^{n+1}}{1 - \frac{3}{4}} = 4\left(1 - \left(\frac{3}{4}\right)^{n+1}\right), \quad n \geq 0$

$y[n] = 0, \quad n < 0$

d) as $n \to \infty \quad y[n] \longrightarrow 4$

5. BIBO stability proof for an LTI system (ECEN 5632 problem)

Assume $|x[n]| \le B_x$ for all n and that $\sum_k |h[k]| < \infty$. Starting from $y[n] = \sum_k x[k]h[n-k]$, prove that $y[n]$ is bounded for all n. State an explicit bound B_y.

Then explain in one sentence why $h[n] = (0.99)^n\, u[n]$ is stable even though its impulse response lasts forever.

$$|x[n]| \leq B_x$$

$$|y[n]| \leq \sum_k |x[k]|\,|h[n-k]|$$ by the triangle inequality

$$\leq B_x \sum_k |h[n-k]|$$

$$= B_x \sum_m |h[m]| < \infty$$ if $\sum_k |h[k]| < \infty$

For $h[n] = (0.99)^n u[n]$ the absolute sum is

$$\sum_{n=0}^{\infty} |0.99|^n = \frac{1}{1-0.99} = 100 < \infty \Rightarrow \text{stable}$$